Introduction

Why Study Hardware

- Decline of Moore's Law
- Proliferation of multi-core processors
- Emergence of new platforms

Hardware knowledge helps programmers write better code

This is a first step for chip/OS/compiler developers

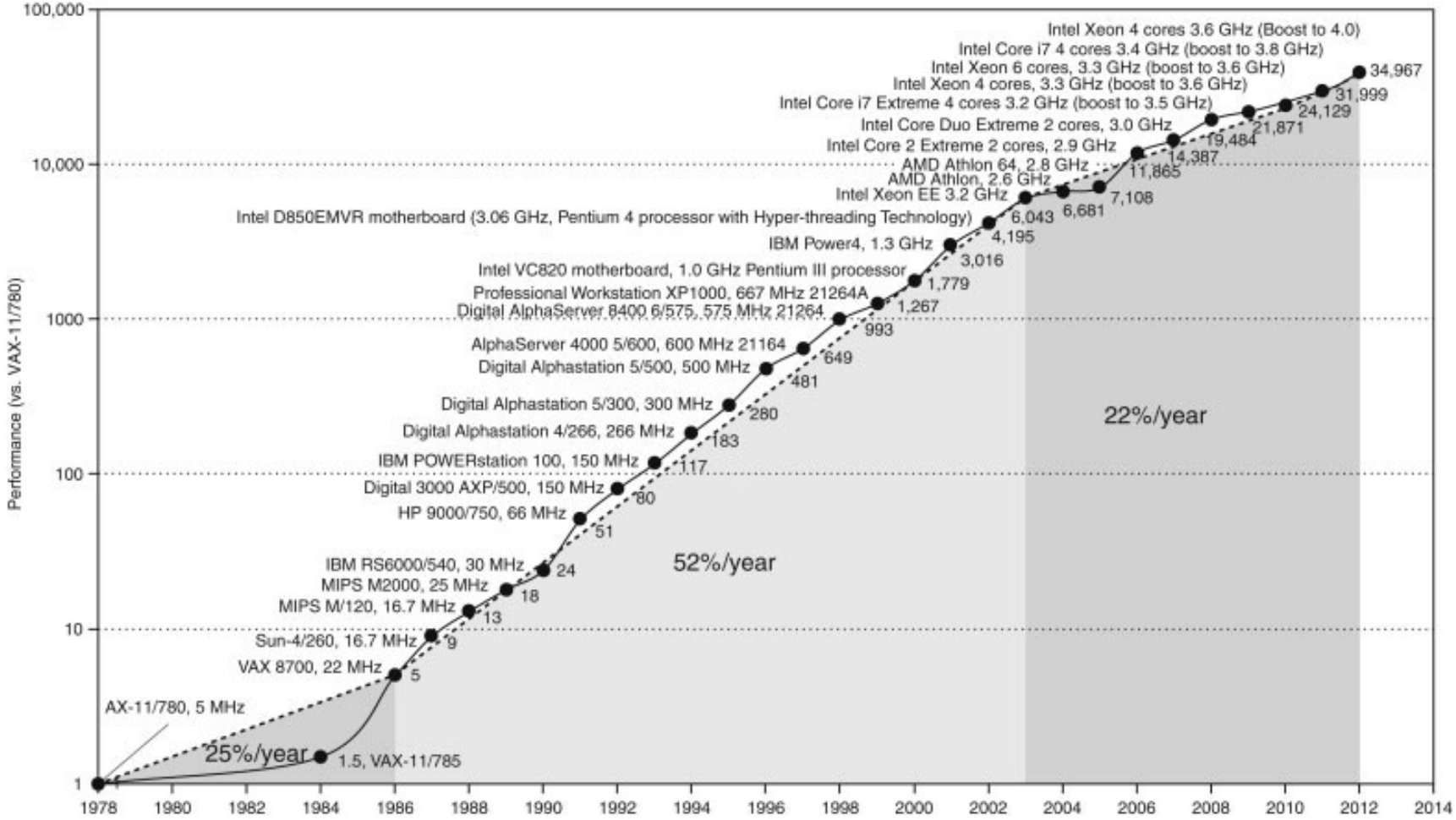
Example

200x speedup for matrix vector multiplication

- Data level parallelism: 3.8x
- Loop unrolling and out-of-order execution: 2.3x
- Cache blocking: 2.5x
- Thread level parallelism: 14x

David Patterson and John Hennessy. Computer Organization and Design -- The Hardware/Software Interface.

Microprocessor Performance



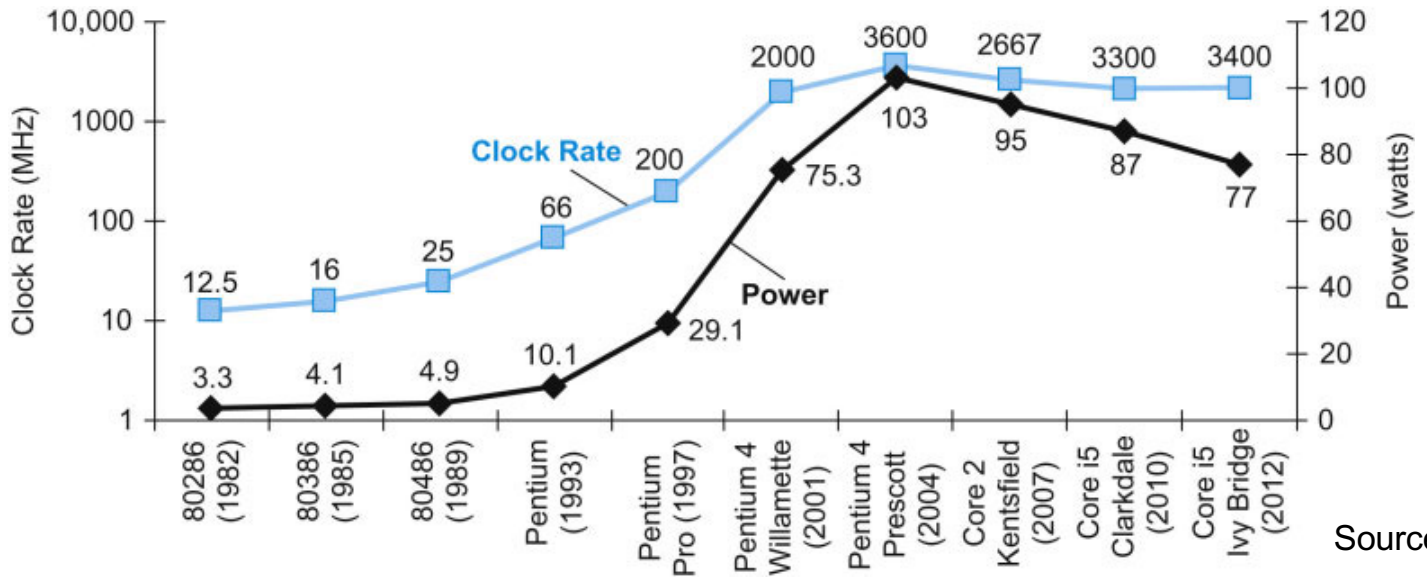
Source: H&P Textbook

50% improvement every year!!

What contributes to this improvement?

Power Consumption Trends

- Dyn power α activity x capacitance x voltage^2 x frequency
- Voltage and frequency are somewhat constant now, while capacitance per transistor is decreasing and number of transistors (activity) is increasing
- Leakage power is also rising (function of #trans and voltage)



Source: H&P Textbook

A Primer on Clocks and Cycles

Performance Equation - I

CPU execution time = CPU clock cycles x Clock cycle time
Clock cycle time = 1 / Clock speed

If a processor has a frequency of 3 GHz, the clock ticks 3 billion times in a second – as we'll soon see, with each clock tick, one or more/less instructions may complete

If a program runs for 10 seconds on a 3 GHz processor, how many clock cycles did it run for?

If a program runs for 2 billion clock cycles on a 1.5 GHz processor, what is the execution time in seconds?

MIPS Instruction Set

A Basic MIPS Instruction

C code: a = b + c ;

Assembly code: (human-friendly machine instructions)

```
add   a, b, c     #  a is the sum of b and c
```

Machine code: (hardware-friendly machine instructions)

00000010001100100100000000100000

Translate the following C code into assembly code:

a = b + c + d + e;

Instruction Set

- Important design principles when defining the instruction set architecture (ISA):
  - keep the hardware simple – the chip must only implement basic primitives and run fast
  - keep the instructions regular – simplifies the decoding/scheduling of instructions

We will later discuss RISC vs CISC

Example

C code a = b + c + d + e;

translates into the following assembly code:

```
add  a, b, c          add  a, b, c
add  a, a, d    or    add  f, d, e
add  a, a, e          add  a, a, f
```

- Instructions are simple: fixed number of operands (unlike C)
- A single line of C code is converted into multiple lines of assembly code
- Some sequences are better than others… the second sequence needs one more (temporary) variable f

Subtract Example

C code f = (g + h) – (i + j);

translates into the following assembly code:

```
add  t0, g, h          add  f, g, h
add  t1,  i, j    or   sub   f, f, i
sub  f,   t0, t1       sub   f, f, j
```

- Each version may produce a different result because floating-point operations are not necessarily associative and commutative… more on this later

Operands

- In C, each “variable” is a location in memory
- In hardware, each memory access is expensive – if variable *a* is accessed repeatedly, it helps to bring the variable into an on-chip scratchpad and operate on the scratchpad (registers)
- To simplify the instructions, we require that each instruction (add, sub) only operate on registers
- Note: the number of operands (variables) in a C program is very large; the number of operands in assembly is fixed… there can be only so many scratchpad registers

Registers

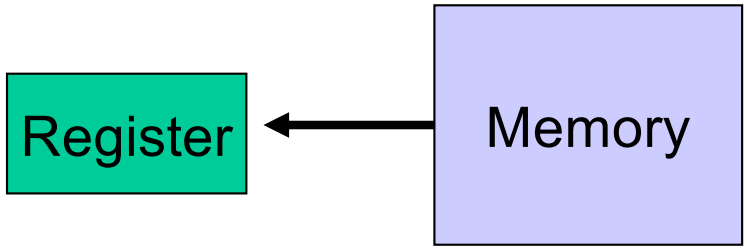
- The MIPS ISA has 32 registers (x86 has 8 registers) – Why not more? Why not less?
- Each register is 32 bits wide (modern 64-bit architectures have 64-bit wide registers)
- A 32-bit entity (4 bytes) is referred to as a word
- To make the code more readable, registers are partitioned as $s0-$s7 (C/Java variables), $t0-$t9 (temporary variables)…

```
add  $s0, $s1, $s2
```

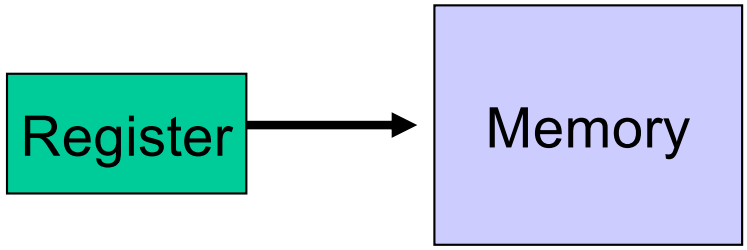
Memory Operands

- Values must be fetched from memory before (add and sub) instructions can operate on them

Load word
lw $t0, memory-address



Store word
sw $t0, memory-address



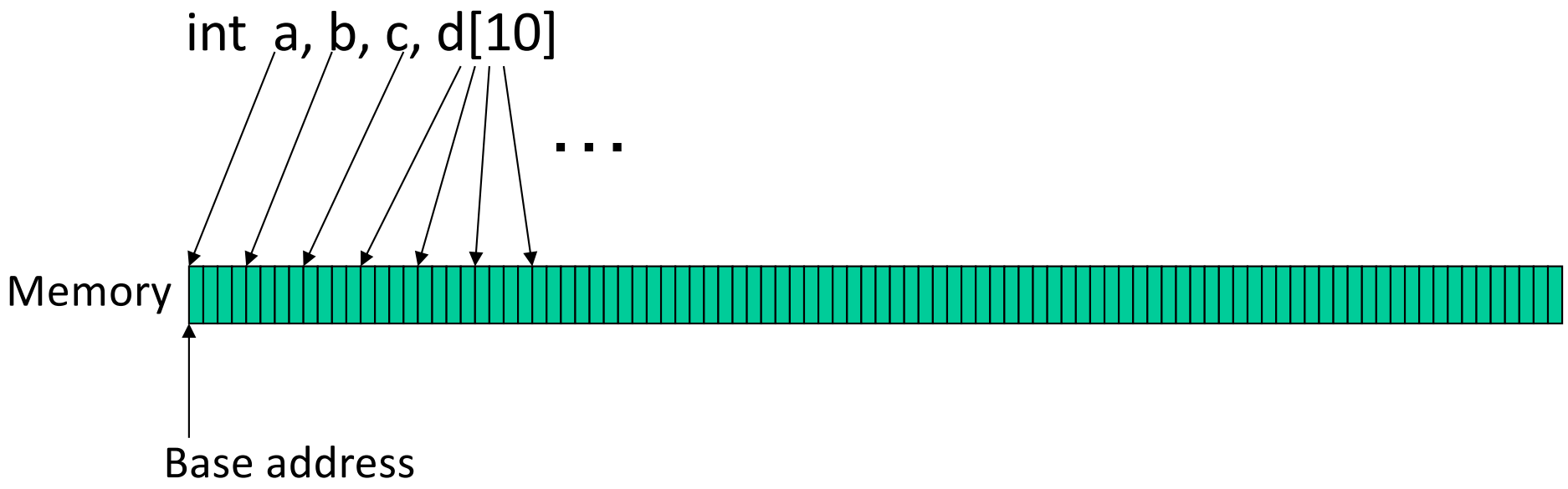
How is memory-address determined?

Binary Stuff

- 8 bits = 1 Byte, also written as 8b = 1B
- 1 word = 32 bits = 4B
- 1KB = 1024 B = 2^{10} B
- 1MB = 1024 x 1024 B = 2^{20} B
- 1GB = 1024 x 1024 x 1024 B = 2^{30} B
- A 32-bit memory address refers to a number between 0 and $2^{32} - 1$, i.e., it identifies a byte in a 4GB memory

Memory Address

- The compiler organizes data in memory… it knows the location of every variable (saved in a table)… it can fill in the appropriate mem-address for load-store instructions



Immediate Operands

- An instruction may require a constant as input

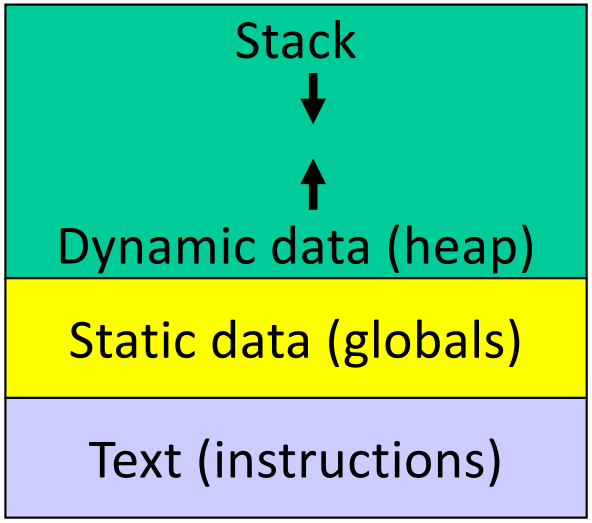
- An immediate instruction uses a constant number as one of the inputs (instead of a register operand)

- Putting a constant in a register requires addition to register $zero (a special register that always has zero in it) -- since every instruction requires at least one operand to be a register

- For example, putting the constant 1000 into a register:
addi $s0, $zero, 1000

Memory Organization

- The space allocated on stack by a procedure is termed the activation record (includes saved values and data local to the procedure) – frame pointer points to the start of the record and stack pointer points to the end – variable addresses are specified relative to $fp as $sp may change during the execution of the procedure
- $gp points to area in memory that saves global variables
- Dynamically allocated storage (with malloc()) is placed on the heap



Base Address and Offsets

C code: a = b + c;

```
addi   $gp, $zero, 1000      # putting base address 1000 into
                             # the global pointer
lw $s2, 4($gp)               # loading variable b into $s2
lw $s3, 8($gp)               # loading variable c into $s3
add $s1, $s2, $s3            # sum in $s1
sw $s1, $gp                  # storing sum into variable a
addi $s4, $gp, 12            # $s4 now contains the start
                             # address of array d[ ]
```

Example

Convert to assembly:

C code: d[3] = d[2] + a;

Assembly

```
lw $t0, 8($s4)        # d[2] is brought into $t0
add $t0, $t0, $s1     # the sum is in $t0
sw $t0, 12($s4)       # $t0 is stored into d[3]
```

Assembly version of the code continues to expand!

Recap – Numeric Representations

- Decimal $35_{10} = 3 \times 10^1 + 5 \times 10^0$
- Binary $00100011_2 = 1 \times 2^5 + 1 \times 2^1 + 1 \times 2^0$
- Hexadecimal (compact representation)

  $0x\ 23$ or $23_{hex} = 2 \times 16^1 + 3 \times 16^0$

  0-15 (decimal) → 0-9, a-f (hex)

Dec	Binary	Hex	Dec	Binary	Hex	Dec	Binary	Hex	Dec	Binary	Hex
0	0000	00	4	0100	04	8	1000	08	12	1100	0c
1	0001	01	5	0101	05	9	1001	09	13	1101	0d
2	0010	02	6	0110	06	10	1010	0a	14	1110	0e
3	0011	03	7	0111	07	11	1011	0b	15	1111	0f